

Problem

If $R = 50\ \Omega$, $L = 1.5\ \text{H}$, what value of C will make an RLC series circuit:

(a) overdamped,

(b) critically damped,

(c) underdamped?

Step-by-step solution

Step 1 of 3

(a)

Calculate the value of capacitance when the circuit is over damped.

Write the condition for over damped circuit.

$$C > \frac{4L}{R^2} \dots\dots (1)$$

Substitute the $50\ \Omega$ for R and $1.5\ \text{H}$ for L in equation (1).

$$\begin{aligned} C &> \frac{4L}{R^2} \\ &> \frac{(4)(1.5)}{(50^2)} \\ &> 2.4\ \text{mF} \end{aligned}$$

Hence, the value of capacitance C must be $> 2.4\ \text{mF}$ for the circuit to be over damped.

Step 2 of 3

(b)

Calculate the value of capacitance when the circuit is critically damped.

Write the condition for critically damped circuit.

$$C = \frac{4L}{R^2} \dots\dots (2)$$

Substitute the $50\ \Omega$ for R and $1.5\ \text{H}$ for L in equation (2).

$$\begin{aligned} C &= \frac{4L}{R^2} \\ &= \frac{(4)(1.5)}{(50^2)} \\ &= 2.4\ \text{mF} \end{aligned}$$

Hence, the value of capacitance C must be $2.4\ \text{mF}$ for the circuit to be critically damped.

Step 3 of 3

(c)

Calculate the value of capacitance when the circuit is under damped.

Write the condition for under damped circuit.

$$C < \frac{4L}{R^2} \dots\dots (3)$$

Substitute the $50 \, \Omega$ for R and $1.5 \, \text{H}$ for L in equation (3).

$$\begin{aligned} C &< \frac{4L}{R^2} \\ &< \frac{(4)(1.5)}{(50^2)} \\ &< 2.4 \, \text{mF} \end{aligned}$$

Hence, the value of capacitance C must be $< 2.4 \, \text{mF}$ for the circuit to be under damped.